

1. Introduction

This project is about developing a VUT Cafeteria Online Food Ordering System for the campus that makes it easier for students, staff, and visitors to order food at the cafeteria online, track their orders, and collect their food when it's ready. Instead of standing in long queues or dealing with delays that waste valuable time for students or staff members in the university. Users will be able to register and login as students or staff, browse available menus, place their orders, and make payments directly from their devices. This approach can help reduce congestion at the cafeteria and improve the overall efficiency of food services on campus.

The main objective of this project is to create a simple, reliable, and user-friendly platform that meets the needs of both customers and food service providers.

2. PROJECT SCOPE

2.1 Project Overview

The project scope defines the features and deliverables included in the development of the VUT Cafeteria Online Food Ordering System. It also identifies the boundaries of the project by outlining what will and will not be included, helping to keep the project focused and prevent scope creep.

The system aims to improve the efficiency of campus cafeteria services by allowing students to place food orders online instead of waiting in long queues. Students will be able to view menus, place orders, and track their order status, while cafeteria staff will manage orders through a vendor dashboard. This system is expected to make the food ordering process faster and more convenient for both students and vendors.

2.2 Project Objectives

The main objectives of this project are to:

- Develop a functional VUT Cafeteria Online Food Ordering System that can be used by students and cafeteria vendors.
- Provide a secure registration and login system that allows users to access the platform safely.
- Allow students to browse cafeteria menus and place food orders online.
- Provide cafeteria vendors with a dashboard to view and manage incoming orders.
- Enable real-time order tracking so students can see the progress of their orders.
- Improve the efficiency of cafeteria services reducing queues and waiting times.

2.3 In Scope Deliverables

The following features and deliverables are included in the scope of this project.

Customer or Student Deliverables

- A student registration and login system
- A menu browsing interface that displays available cafeteria food items
- Online order placement functionality
- Order confirmation notifications after an order has been placed
- Real-time order tracking showing order
- An order history feature that allows students to view their previous orders

food service providers Deliverables

- A service provider's login and authentication system
- A service provider dashboard for viewing incoming customer orders
- Order management tools to accept, prepare, and complete orders
- An order status update feature that allows service providers to update order progress
- A basic reporting interface to view daily orders

System Deliverables

- A database for storing user accounts, menu items, and order information
- A web-based platform that can be accessed by students and vendors through a web browser
- Administrative functionality for managing menu items and system data
- An order tracking system that updates order status for customers

2.4 Out of Scope

The following features will not be included in the VUT Cafeteria Online Food Ordering System:

- Integration with external restaurants or third-party food delivery services
- GPS-based delivery tracking
- Mobile application development, as the system will only be web-based

- Residence or student accommodation delivery services

3. Work Breakdown Structure of Online Food Ordering System

1. Project Initiation

- 1.1 Identify stakeholders
- 1.2 Define project objectives
- 1.3 Develop project charter

2. Project Planning

- 2.1 Define project scope
- 2.2 Develop WBS
- 2.3 Create project schedule
- 2.4 Estimate budget and risks

3. System Analysis

- 3.1 Gather user requirements
- 3.2 Analyze system requirements
- 3.3 Prepare documentation for system requirements

4. System Design

- 4.1 Design system architecture
- 4.2 Design database
- 4.3 Design user interface

5. System Development

- 5.1 Develop front-end (student interface)
- 5.2 Develop back-end (database & logic)
- 5.3 Develop vendor dashboard

6. Testing

- 6.1 Perform unit and integration testing
- 6.2 Conduct user acceptance testing

7. Deployment

- 7.1 Deploy system
- 7.2 Train users

8. Project Closure

- 8.1 Final system review
- 8.2 Documentation and project closure

4. Project Schedule

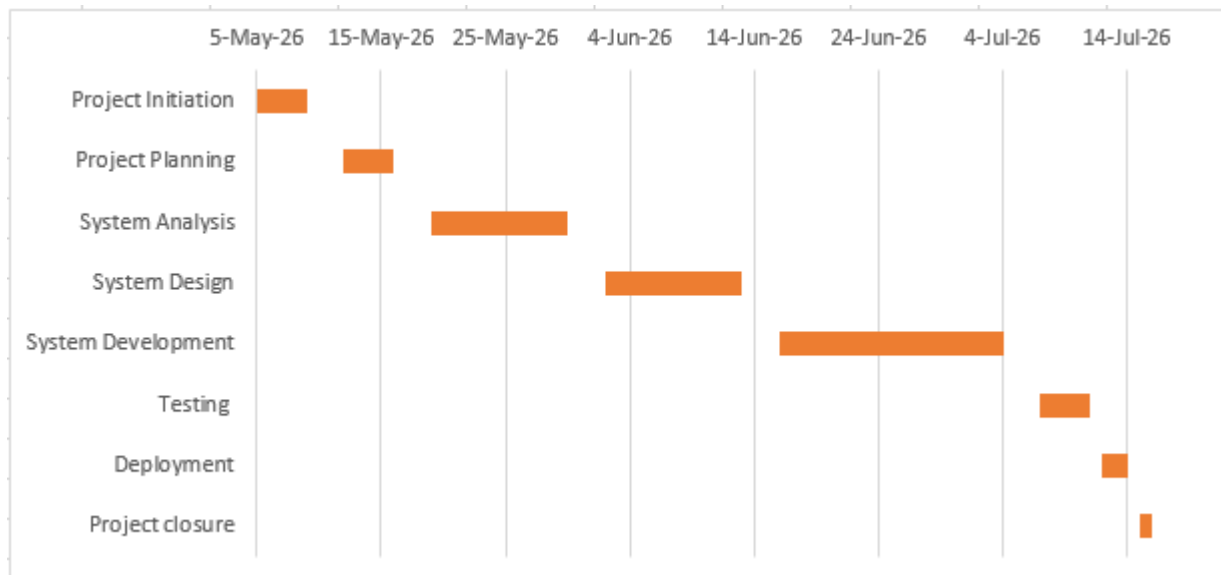
The project schedule outlines the sequence of activities, their durations, and dependencies to successfully complete the VUT Online Food Ordering System. The

schedule ensures that tasks are completed in the correct order and within the allocated timeframe. The total estimated project duration is 11 weeks.

4.1. Project Schedule Table

Task ID	Task Name	Start Date	End Date	Duration	Dependency
1	Project Initiation	5-May-26	9-May-26	4	
2	Project Planning	12-May-26	16-May-26	4	1
3	System Analysis	19-May-26	30-May-26	11	2
4	System Design	2-Jun-26	13-Jun-26	11	3
5	System Development	16-Jun-26	4-Jul-26	18	4
6	Testing	7-Jul-26	11-Jul-26	4	5
7	Deployment	12-Jul-26	14-Jul-26	2	6
8	Project closure	15-Jul-26	16-Jul-26	1	7

4.2 Gantt Chart



5. RESOURCE PLAN

The resource plan outlines the human, hardware, and software resources required to develop and implement the VUT Cafeteria Online Food Ordering System across all project phases.

5.1 Human Resources

- Project Managers: Organize and supervise the project
- Software Developers: Create the front-end and back-end of the system
- Database Administrator: Designs and manages the database
- UI/UX Designer: Designs user-friendly interfaces for students

- System Analyst: Collects requirements and defines system specifications
- Testers (QA Team): Test the system for errors and performance issues
- Network Administrator: Ensures system connectivity and server reliability

5.2 Hardware and Technology Equipment

- Computers and laptops for development and administration
- Smartphones for mobile compatibility testing
- Servers for hosting the system or cloud services
- Internet access (campus network or Wi-Fi)
- Backup storage devices (external drives or cloud storage)

5.3 Software and Other Resources

- Programming tools such as NetBeans and VS Code
- Database software such as MySQL
- Web technologies: HTML, CSS, JavaScript
- Version control systems such as Git
- Documentation tools such as Microsoft Word and PDF tools
- Testing tools for debugging and performance testing
- Payment integration systems

6. QUALITY PLAN

The quality plan ensures that the system meets the required standards of performance, reliable usability in the VUT Cafeteria Online Food Ordering System project.

6.1 Quality Standards

- The system must allow users to register, log in, browse menus, and place food orders successfully
- User interface must be simple and easy to use
- Pages must load quickly and orders must be processed efficiently
- System must operate with minimal downtime
- User data must be protected

- System must work on different devices and browsers
- All data such as prices and orders must be accurate

6.2 Quality Measurement

- Unit, integration and system testing
- Measure page load and order processing time
- Track number of bugs and system errors
- Collect feedback from users
- Measure successful vs failed transactions

6.3 Quality Monitoring and Control

- Continuous testing after deployment
- Use performance monitoring tools
- Provide user support and feedback systems
- Regular system updates and maintenance
- Security monitoring and vulnerability checks

7. RISK MANAGEMENT PLAN

Risk	Impact	Probability	Mitigation Strategy
Vendor resistance to adoption	High	Medium	Early stakeholder engagement and pilot incentives
Low student adoption	High	Medium	Marketing campaign and promotional offers
Payment integration delays	Medium	Medium	Parallel development and backup provider
Network infrastructure issues	Medium	Low	Cloud-based solution with offline mode
Scope creep	Medium	High	Strict change control process

Risk Monitoring

Risks will be reviewed weekly during project meetings. A risk register will be maintained and updated throughout the project. Escalation procedures are in place for high-impact risks requiring senior management intervention.

8. COMMUNICATION PLAN

Activity	Frequency	Audience	Purpose
Daily Standup	Daily	Project Team	Progress updates
Sprint Planning	Weekly	Development Team	Task allocation
Stakeholder Meeting	Monthly	Sponsors, Vendors	Status reporting
Progress Report	Weekly	Project Sponsor	Written updates
User Feedback Session	Weekly	Students	Requirements validation

Communication Channels

- The use of e-mail for formal communications and documentation
- WhatsApp for daily collaboration and quick updates
- Google docs for task tracking, technical discussions and for document management

9. Project Budget

9.1 Cost Estimation Approach

The budget was created using a bottom-up approach and both direct costs and indirect costs were considered. A contingency amount was also added to cover unexpected costs.

9.2 Estimated Project Costs

Category	Description	Cost (In Rands)
Personnel Costs		
Business Analyst	Gathering and analysing requirements	25 000
System Designer	Designing system and user interface	20 000
Developer	System development (front-end & back-end)	45 000
Tester	Testing the system	15 000
Project Manager	Managing and monitoring the project	20 000
SUBTOTAL		125 000
Technology Costs		
Web Hosting	Hosting the system online	3 000
Domain Name	Website registration	500
Development tools	Software and tools	5 000
Database Setup	Database configuration	4 000
SUBTOTAL		12 500
Training & Deployment	Training users and system setup	8 000
Miscellaneous Costs	Admin , communication , documentation	5 000
Contingency (10%)	Extra costs for unexpected issues	15 050
TOTAL BUDGET		R165 550

9.3 Cost Baseline

The estimated R165 550 will be used as the cost baseline to measure and control spending during the project.

9.4 Budget Control

To make sure the project stays within budget:

- Actual costs will be checked against planned costs
- Any overspending will be identified early
- Changes that affect the budget must be approved
- Regular monitoring will be done throughout the project

9.5 Assumptions

The following assumptions were made :

- The project will take about 11 weeks to complete
- Most tools used will be affordable and free
- The team is small and some members may handle more than one role
- No expensive hardware is required

10. Conclusion

In conclusion, this project plan outlines the development of the VUT Cafeteria Online Food Ordering System, which aims to improve the efficiency of food services on campus. The system is designed to reduce long queues, save time for students and staff , and provide a more convenient way of ordering food Online

The project scope clearly defines what will be included in the system , such as user registration , menu browsing , online ordering , and order tracking .The Work Breakdown Structure (WBS) and project schedule provides a clear structure of the tasks and timelines needed to complete the project within the estimated 11 weeks . In addition , the communication plan ensures that all stakeholders are kept informed throughout the project , while the project budget provides a realistic estimate of the financial resources required. Proper cost management and monitoring will help ensure that the project stays within budget. Overall, this project plan provides a clear roadmap for successfully developing and implementing the system. If all phases are properly executed, the system will meet its objectives and improve the overall cafeteria experience at VUT.